Semester 5 · Mechanical / Manufacturing · Program Core · 4 Credits

Refrigeration and Air-Conditioning

Handbook for refrigeration principles, vapour compression and absorption systems, refrigerants, refrigeration equipment, cryogenics, psychrometry, air conditioning and cooling load estimation.

Course meaning

Refrigeration removes heat from a low-temperature region and rejects it to a high-temperature region by using work or heat input. Air-conditioning controls temperature, humidity, purity and air motion for comfort or industrial requirement.

Malayalam: Cooling എന്നത് cold air ഉണ്ടാക്കൽ മാത്രം അല്ല. Heat remove ചെയ്യുക, humidity control ചെയ്യുക, compressor/condenser/expansion/evaporator cycle മനസ്സിലാക്കുക എന്നതാണ്.

Official details

Code	5022
Title	Refrigeration and Air-Conditioning
Category	Program Core
Prerequisites	Mathematics I and II, Thermal Engineering

CO1

Refrigeration cycles and performance parameters.

CO2

VCR, VAR systems and refrigerants.

CO3

Components and applications of refrigeration.

CO4

Psychrometry, AC systems and cooling load.

Roadmap

CO1 · 8 Hours

Refrigeration Cycles and COP

Basic terms

Refrigerating effect is the heat removed from the cooled space. Ton of refrigeration is the rate of heat removal equivalent to freezing one ton of water at 0 degree C into ice at 0 degree C in 24 hours. COP is refrigerating effect divided by work input.

COP = Refrigerating effect / Work input.

1 TR ≈ 3.517 kW.

Cycles

Reversed Carnot cycle is ideal. Air refrigeration uses air as working medium and includes open and closed cycles. Bell-Coleman cycle is reversed Brayton cycle. Air refrigeration is simple and safe but has lower COP and larger equipment.

Worked example

If plant capacity is 2 TR, refrigerating effect = 2 × 3.517 = 7.034 kW.

Cooling capacity = 7.034 kW.

Expected questions

1. Define refrigeration, refrigerating effect, TR and COP.
2. Explain reversed Carnot and Bell-Coleman cycle.
3. List advantages and disadvantages of air refrigeration.

CO2 · 13 Hours

Vapour Compression, Absorption and Refrigerants

Vapour compression system

Basic components are compressor, condenser, expansion device and evaporator. The compressor raises pressure; condenser rejects heat; expansion device reduces pressure; evaporator absorbs heat. P-H and T-S diagrams show work, heat rejection and refrigerating effect.

Cycle effects

Suction pressure, discharge pressure, superheating and undercooling affect COP and capacity. Undercooling generally increases refrigerating effect. Excessive superheating may increase compressor work.

Vapour absorption and refrigerants

Absorption systems use heat input instead of mechanical compressor work. Refrigerants should have suitable boiling point, high latent heat, low toxicity, non-flammability, chemical stability and low environmental impact. ODP and GWP are critical.

Expected questions

1. Draw VCR flow diagram and explain.
2. Represent VCR on P-H and T-S diagrams.
3. Compare VCR and VAR systems.
4. Explain desirable properties and leakage detection of refrigerants.

CO3 · 15 Hours

Refrigeration Equipment and Cryogenics

Compressors and condensers

Compressors may be reciprocating, rotary roller, rotary vane and centrifugal. Hermetically sealed compressors house motor and compressor together. Condensers may be air-cooled or water-cooled shell and tube, shell and coil or double tube type.

Evaporators and controls

Evaporators may be dry or flooded, natural or forced convection. Flow controls include capillary tube, automatic expansion valve and thermostatic expansion valve.

Applications and cryogenics

Domestic refrigerators, water coolers and ice plants are common applications. Cryogenics deals with very low temperature refrigeration and includes cascade refrigeration, Joule-Thomson effect and liquefaction of nitrogen and hydrogen.

Expected questions

1. Classify compressors with sketches.
2. Explain air-cooled and water-cooled condensers.

3. Explain capillary tube, AXV and TXV.
4. Explain cascade refrigeration and Joule-Thomson effect.

CO4 · 22 Hours

Psychrometry, Air Conditioning and Cooling Load

Psychrometric terms

Psychrometry studies moist air. Important terms are dry bulb temperature, wet bulb temperature, dew point, relative humidity, specific humidity, enthalpy, degree of saturation and Dalton law of partial pressure.

Psychrometric processes

Processes include sensible heating, sensible cooling, humidification, dehumidification, cooling and dehumidification, heating and humidification, bypass factor, sensible heat factor and coil efficiency.

AC systems and load

Air conditioning may be comfort or industrial, unitary or central. Summer, winter and year-round systems control temperature and humidity. Cooling load includes people, lights, equipment, solar heat gain, infiltration and ventilation load.

Expected questions

1. Define DBT, WBT, DPT, RH and specific humidity.
2. Solve simple chart/table problems.
3. Explain summer, winter and year-round AC with line sketches.
4. List sources of heat gain and explain HVAC concept.

Formula bank

COP = RE/W. 1 TR = 3.517 kW. Refrigerating effect in VCR = $h1 - h4$. Compressor work = $h2 - h1$. Relative humidity = actual vapour pressure / saturation vapour pressure.

Model paper

1. Define COP and ton of refrigeration.
2. Explain VCR with P-H and T-S diagrams.
3. Explain compressors, condensers and evaporators.
4. Explain psychrometric processes and cooling load estimation.

MCQ practice

1. Compressor raises refrigerant pressure. 2. Condenser rejects heat. 3. Evaporator absorbs heat. 4. DBT is measured by ordinary thermometer.

Answer guide

For refrigeration numericals write state points, enthalpies, formula, substitution, COP and units. For diagrams label compressor, condenser, expansion valve and evaporator correctly. For AC answers mention comfort conditions, humidity control and heat gain sources.

References: R.S. Khurmi and J.K. Gupta, A.S. Sarao and P.S. Gabi, Sadhu Singh, S. Domkundwar, Roy J. Dossat, C.P. Arora, NPTEL, Danfoss and Bharat Skills.